

Manual

Monitoring of Machine Data (MOC)

MDE-MMD 8.2

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1 Overview of Machine Data Monitoring

Possible fields of application

The function package "Machine Data Monitoring" allows for the data recorded in the system to be evaluated with respect to the current status of machines and workplaces as well as in relation to the posted status durations.

Implementation notes

You use the function package if you wish:

- to monitor your machines and workplaces
- to evaluate the recorded times and durations in relation to machines/workplaces
- to evaluate the hours occurred in a special sector/department over a specific period of time in relation to different statuses

Integration

The machine data monitoring evaluates machines and workplaces. The times recorded in the machine data collection module are generally used for this purpose.

Functions

- Workplace/machine overview
 - Tabular workplace/machine overview to represent the current machine status. Display of a picture of the machine. Detailed views, e.g. list of registered staff, operations, tools or resources (available if relevant functions from these HYDRA areas are used at the same time as MDE (machine data collection)). Comparison of actual and target quantities relating to orders, target and actual cycles as well as target and actual stroke numbers.
- Cycle progression
 - Tabular and graphic evaluation on the actual cycle of a machine over configurable periods; can be displayed in "seconds per piece" or "pieces per minute".
- Machine time profile
 - Graphic machine time profile with real-time presentation of the downtime/production performance of several machines over configurable periods including zoom function.
- Downtime profile
 - Drill-down function within the machine time profile for detailed presentation of the downtime performance of a machine
- Machine history

- Machine history for the tabular presentation of machine-related and order-related postings performed at a machine/workplace.

2 Workplaces/Machines

Overview

Menu	Production Facility Management → Current Information → Workplaces/Machines
Transaction code	wpov
Function authorization	wpov

Purpose

The application *Workplaces/machines* provides an evaluation for the production management. It is intended for the following users: users from production scheduling and monitoring, schedulers, supervisors, operators or all MOC users who would like to get a comprehensive overview of the production situation at specific workplaces/machines or a complete organizational unit.

Integration

The application *Workplaces/machines* provides all kind of information that is relevant for workplaces. In addition to master data, the function also provides data required to control production processes. These are, for example:

- current workplace/machine status
- operations currently running at the workplace/machine
- currently used tools and resources
- cycle progression of the shift (for machines with clocked production)
- output per shift: quantities, durations

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the machine or workplace master data. You can also use wildcards (placeholders *).

Group from ... to ...

This selection criterion refers to the group stored in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected group. You can also use wildcards.

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected report group.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows the machines matching the specified character string. You can also use wildcards (placeholders *).

Short name

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

Responsibility area

This selection criterion refers to the responsibility area stored in the machine master data. Note: The user can only view those machines that are included in the responsibility areas assigned to the user.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected company. You can also use wildcards.

Cost center

This selection criterion refers to the cost center stored in the machine and/or workplace master data. All workplaces/machines assigned to the selected cost center are displayed. You can also use wildcards.

Status

This selection criterion refers to the current status of machines or workplaces. All machines or workplaces, which are currently assigned to the selected status, are displayed.

Status longer than

This selection criterion refers to the current status of machines or workplaces. All machines or workplaces are shown that are currently assigned to the selected status and that are assigned to this status for a longer period than the one specified.

If several selection criteria are used, the application *Workplaces/machines* shows the results that match all selection criteria.

Detail application *Workplaces*

The detail application *Workplaces* displays all workplaces in accordance with the selections made in the selection panel. The application displays the current status, workplace information, shift quantities, cycles and number of strokes. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

In addition operation-related data is shown, if an operation is currently logged on. In case several operations are logged on, only the first operation is shown in the detail application.

Status

The *Status* column summarizes the different statuses and presents them as an "LED". The colors are as follows:

Light green	Status with RPA 11 (normally "Production")
Blue	Status with RPA 7 (normally "Setup")
Red	Status 30000 (normally "Not assigned")
Gray	Status 20000 or status with RPA 12 (normally break/no shift
Yellow	Status < 10000 and RPA <> [7,11,12] other statuses/downtimes

Master data:

Workplace

Unique ID defined in the workplace configuration.

Short name

Machine name as defined in the workplace configuration.

Designation

Long text/comment on the machine as defined in the workplace configuration.

Gruppen

Group the machine is assigned to in the workplace configuration.

Cost center

Cost center as defined in the workplace configuration.

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as defined in the workplace configuration.

Type

Workplace model according to workplace configuration.

Type

Workplace type according to the workplace configuration.

Status**Status**

Status number of the status that is currently active at the workplace. Color of the currently active status according to configuration.

Status name

Status name of the status that is currently active at the workplace.

Status since

Date when the status was assigned.

Status since

Point in time when the status was assigned.

Duration so far

Present duration of the status that is currently active at this workplace.

Predicted duration

Expected duration of the malfunction entered by the employee when assigning the status in the terminal or the duration that is stored in the status configuration.

Expected end

Calculated point in time when the malfunction ends. The calculation is based on the predicted duration. The end time is calculated using the values of *Date* + *Predicted duration*, synchronized with the Gregorian calendar.

Expected remaining runtime

Expected end minus current time, i.e. "now". If the remaining runtime is negative the expected end is already overdue. In this case, the field is highlighted in red.



Do not confuse the expected remaining runtime of the malfunction with the remaining runtime of the operation.

Shift quantities, primary quantity unit/secondary quantity unit/tertiary quantity unit/base quantity unit**Yield**

Yield that has been posted so far at the selected workplace within the current shift.

Scrap

Scrap that has been posted so far at the selected workplace in the current shift.

Rework

Rework quantity that has been posted so far at the selected workplace in the current shift.

Open quantity

Open quantity that has been posted so far at the selected workplace within the current shift.

Unit

Unit of primary quantity

Zyklus**Target cycle**

Current target cycle at the workplace.

If an operation is logged on to the machine the target cycle defined for the operation is displayed in seconds per cycle. There is no target cycle for machines to which no OP is currently logged on. In this case, "0" is entered in the "target cycle" field.

Actual cycle

Current actual cycle of the workplace

Colored display of the actual cycle relating to the configured cycle parameters.

Difference (%)

The difference in% is calculated according to the following formula: (target cycle - actual cycle) / target cycle * 100%. If the actual cycle is slower than the target cycle, the difference is indicated in negative values, otherwise positive values are shown. See below for coloring.

Actual cycle (OP)

The actual cycle (OP) is a value referring to the order. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status.

Formula: Actual cycle_{OP} = RPA11_{OP} / (Yield_{OP} / Partitioning_{OP})

Difference (OP) (%)

The difference OP [%] column is computed according to the following formula:
$$\text{Difference}_{\text{OP}} = \text{Abs}((\text{target cycle number} - \text{actual cycle number}_{\text{OP}}) * 100) / \text{target cycle number}$$

Cycle number [1/min]**Target cycle number**

1 / Target cycle

There is no target cycle for machines to which no OP is currently logged on. For this reason, the target stroke number is 0.

Actual cycle number

1 / actual cycle

Difference (%)

$(\text{target cycle number} - \text{actual cycle number}) / \text{target stroke number} * 100\%$

Please note: For rounding reasons, the difference indicated here might deviate from the difference shown in the "cycle" category.

Actual cycle number (OP)

The actual cycle number (OP) is a value relating to orders. The values used for the calculation all refer to order logons and, as a result, they are independent from the current machine status. Formula: $\text{Actual cycle number}_{OP} = \text{yield}_{OP} / (\text{partitioning}_{OP} * \text{RPA11}_{OP})$

Difference (OP) (%)

The difference OP column is computed by the following formula:

$\text{Difference}_{OP} = \text{Abs}((\text{target cycle number} - \text{actual cycle number}_{OP}) / \text{target cycle number} * 100)$

Coloring of the column *Difference*

In the master data, you can define the coloring of the *Difference* column in the *Cycle* category per machine for the upper/lower action limits or upper/lower tolerance limits (menu: master data > workplaces/machines > cycle parameters). The value with a sign showing the difference is used for coloring. The value in the difference column is displayed in red if the tolerance limits are exceeded; the value is displayed in blue if the action limits are exceeded. The data is not displayed in color if no cycle parameters are defined.

Order quantities

This category displays data relating to quantities of the operation currently logged on. In case several operations are logged on, only the first operation is shown in the detail application.

Target quantity (P)

Target quantity (primary quantity unit) of the operation currently logged on.

Target scrap (P)

Target scrap (primary quantity unit) of the operation currently logged on.

Yield (P)

Posted yield (primary quantity unit) of the operation currently logged on.

Scrap (P)

Posted scrap (primary quantity unit) of the operation currently logged on.

Rework (P)

Posted rework (primary quantity unit) of the operation currently logged on.

Open quantity (P)

Posted open quantity (primary quantity unit) of the operation currently logged on.

Total quantity (P)

Total of yield + scrap + rework + outstanding quantity (open quantity)

Unit (P)

Unit of the primary quantity unit of the operation currently logged on.

Difference [%]

This difference identifies the percentage that is still to be produced to reach the target quantity of the operation. To this end, the already posted yield (P) is set in ratio to the target quantity (P):

$$\text{Difference} = 100 - (100 / \text{target quantity (P)} * \text{yield (P)})$$

The result is displayed with 2 decimal places.

Times relating to operations

This category displays the durations which are posted to the individual resource performance accounts of the logged on operation.

Detail application *Image*

The picture in the *Image* detail application shows the picture of the machine as stored in the machine configuration. The image of the machine selected in the detail application “workplace” is displayed.

The following image formats are supported: jpg, gif, png, tif, bmp, ico, emf, and wmf. The pictures have to be filed in a directory that may be accessed via the path ID “MOCWPIMG” within the path configuration.



Further information on the configuration can be found [here](#).

Detail application *Operations logged on*

The detail application *Operations logged on* shows all operations that are currently logged on to workplaces/machines which are selected in the detail application “Workplaces”. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the operation is logged on.

Order

Order

Order number of the operation.

Sequence

Sequence number of the OP (if sequences are used).

OP

Operation number

Split

Split number of the operation (if the split function is used).

SOP

Sub operation number (reserved).

OP name/designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data.

Login

Date

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

Primary quantity/secondary quantity/tertiary quantity/base quantity

Target quantity

Target quantity of the operation

Unit

Unit of primary quantity

Yield

Yield that has been posted so far to the operation

Scrap

Scrap that has been posted so far to the operation

Rework

Rework quantity that has been posted so far to the operation

Open quantity

Open quantity that has been posted so far to the operation

Yield/target quantity [%]

Proportion of yield to target quantity in %

Yield since logon

Yield since the operation is logged on

Detail application *Staff logged on*

The detail application *Staff logged on* shows all persons who are logged on to the workplace selected in the detail application "Workplace". The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace where the operation is logged on.

Person**Name**

The person's name as defined in the HR master.

First name

The person's first name as defined in the HR master.

Name

The person's complete name as defined in the HR master (last name, middle name and first name)

Company

Company the person is assigned to in the HR master.

Personnel number

Unique key to identify the person. (Key)

Staff badge number

Staff badge number assigned to this person in the HR master.

Operator position/function

Abbreviation of the operator's function ("operator position") that has been selected when the person logged on to the machine.

Operator position/function

Unique key of the operator position that has been selected when the person logged on to this machine.

Order

Order

Order number of the operation.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (provided that the split function is used).

SOP

Sub operation number (reserved).

OP name/designation

Designation of the operation

Article

Article number produced by the operation; taken over from operation data.

Login

Date

Date when the operation was last logged on to this workplace

Time

Time when the operation was last logged on to this workplace

"Advance logon" option

If this option is set, the person is logged on automatically when shifts change the next time.

Detail application *Resources logged on*

The detail application *Resources logged on* shows all resources which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace

Workplace

Workplace where the operation is logged on.

Resource

Resource type

Resource type to which the resource is assigned.

Resource

Resource ID that is entered in the resource master data.

Designation

Resource designation recorded within master data.

Resource family

Resource family (internal ID) to which the resource is assigned.

Login

Date

Date when the resource was last logged on to this workplace.

Time

Time when the resource was last logged on to this workplace.

Detail application *Maintenance*

The detail application *Maintenance* shows all active maintenances for the workplace that is currently selected in the selection panel. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Maintenance

Active

Light green: Active

Status

Status of maintenance activity

Green

Blue "blue" threshold has been exceeded

Yellow "yellow" threshold has been exceeded

Red "red" threshold has been exceeded

Maintenance

Maintenance name

Type

Maintenance type defined for the maintenance:

- T (cycle-based)
- B (operating hours)
- Z (time-based)

Class

Maintenance class

Non-recurring maintenance

Flag indicating that this maintenance is only performed once.

Valid from

Start of maintenance validity. A maintenance can only fall due within the validity period.

Valid until

End of maintenance validity.

Maintenance order

Maintenance order assigned to this maintenance.

Date

Date when this maintenance was last carried out at the selected machine.

Time

Time when this maintenance was last carried out at the selected machine.

Modified by

Person (user) who reset the last maintenance.

Actual cycles

Number of cycles accrued so far.

Next maintenance after

Counter reading of cycles when the next maintenance is to be performed.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Actual duration

Operating hours, which have been posted so far onto the resource – according to resource type settings.

Next maintenance after

Meter reading of the operating hours counter triggering the next maintenance to become due.

Interval

Interval in hours within which the maintenance is to be performed; from the maintenance configuration.

Next maintenance on

Date when the next maintenance falls due.

Interval

Interval within which the maintenance is to be performed; from the maintenance configuration.

Info 1 - 6

Additional text 1-6 from the maintenance configuration

Detail application *Article in production*

The detail application *Article in production* shows all output materials with the relevant batch number which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace**Workplace**

Workplace to which the batch is logged on.

Material**Material**

Material number of the currently produced article

Material designation/name

Material name of the currently produced article, which is taken over from the producing operation.

Material type

Material type of the currently produced article, which is taken over from the producing operation.

Batch number

Current batch numbers produced by the OP using this article.

Quantities**Quantities**

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Login

Date

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who performed the last output batch change.

Detail application *Material in use*

The detail application *Material in use* shows all input materials which are logged on to the workplace selected in the detail application *Workplace*. The following paragraphs describe the data available in the table. This data might not be displayed by default. Use the column selection function to add the required data.

Workplace

Workplace

Workplace where the input batch is logged on.

Material

Material

Material number of the currently logged on input batch.

Material designation/name

Material name of the currently logged on material, which is taken over from the producing operation.

Material type

Material type of the currently logged on material, which is taken over from the producing OP.

Batch number

Current batch number of the currently logged on input batch.

Quantities

Original quantity of the batch

Remaining quantity

Remaining quantity of the batch

Quantity unit

Quantity unit in which the batch is managed.

Login

Date

Date when the batch was last logged on to this workplace.

Time

Time when the batch was last logged on to this workplace.

Person

Person (personnel number) who has performed the last input batch logon.

Detail application *Status*

The detail application *Status* shows the current machine status and in parallel all current resource statuses.



Parallel resource statuses refer to a workplace or a machine. The statuses do not depend on the WRM resource statuses.

The entry of other statuses than the machine status requires additional licenses and configurations.

The following paragraphs describe the data available in the table. Some data might not be displayed by default. Use the column selection function to add the required data.

Status

Status

Number of the parallel [Resource status](#).

Status text

[Designation](#) of the parallel resource status

Status type

Number of the [Status type](#) the resource status is assigned to.

Login

Beginning of status

Point in time when the status was set.

Duration

Duration since beginning of status. The duration is calculated based on the Gregorian calendar.

The displayed duration of the status type "MST" (machine status) can therefore differ from the value *Duration so far* in the detail application *Workplaces*.

Resource

Resource type key

Resource type of the workplace/the machine - always "MNR"

Resource

Number of the workplace/machine

Detail application *Shift times*

The detail application *Shift times* shows RPA times of the current shift at the workplace selected in the detail application *Workplace* in a pie chart.

Detail application *Shift quantities*

The detail application *Shift quantities* shows the current shift quantities in a bar chart, i.e. yield, scrap in primary quantity unit. The quantities refer to the workplace selected in the detail application *Workplace*.

Detail application *Cycle progression*

The detail application *Cycle progression* shows the stored cycle values in a line chart in [sec/cycle]. The chart displays the cycle progression of the workplace selected in the detail application *Workplace*. By clicking a radio button the user can decide whether they want to display the current shift or the last x hours. However, x should be less than 8 hours for performance reasons.

The following limit values are displayed as lines: upper tolerance limit - UTL (red), lower tolerance limit - LTL (red), upper action limit - UAL (yellow), lower action limit - LAL (yellow). The limits are computed and displayed on the basis of the *Process parameters* configuration.

Please note: The display depends essentially on the size of the detail application.

Detail application *Downtime ranking list*

The *Downtime ranking list* shows the top x of current downtimes (status is not *production*) of the currently selected workplace during the current shift or the last hours. They are represented in a horizontal bar chart.

Using the radio buttons, it is possible to show the statuses, which have so far occurred in the current shift, or the statuses of the last x hours. By another radio button, the user can configure the display according to downtime durations or the number of respective downtimes.

The TOP X input field allows for the number of statuses to be defined (preassignment: 5).

The color of status bars corresponds to the color defined for the status text within the HYDRA configuration. The status bar is displayed in gray, in case no color is defined for the status. The status text and the value (duration in hours or number) are displayed for each bar.

Toolbar

Data collection



Log on

Use the [Log on](#) function to log on operations to the system.



Partial confirmation

Use the function "[Partial confirmation](#)" to enter part quantities for operations that are then recorded in the system.



Interrupt

Use the function "[Interrupt](#)" to interrupt operations.



Log off

Use the [Log off](#) function to log off operations.



Terminate

Interrupted or prepared operations can be logged off from the system using the [Terminate](#) function

Persons



Log person on

You can log on a person to an operation/machine using the [Log person on](#) function



Log person off

You can log off a person from the relevant operation/machine using the [Log person off](#) function

3 Machine history

Overview

Menu	Production facility/Resource management → Resource analysis → Machine history
Transaction code	wphi
Function authorization	wphi

Purpose

The machine history is a report for the production management. The application allows for tracking and tracing of events that need to be posted at workplaces in MES. In this context, posting events such as status changes, order, tool, and personnel postings, maintenance activities as well as measures recorded at a workplace are listed in chronological order in a table. You can use various selection criteria to evaluate events.

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace stored in the machine or workplace master data. You can also use wildcards (placeholders *).

Group from ... to ...

This selection criterion refers to the group stored in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected group. You can also use wildcards.

Short name

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

Designation

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows the machines matching the specified character string. You can also use wildcards (placeholders *).

Cost center

This selection criterion refers to the cost center stored in the machine and/or workplace master data. The application shows all machines and/or workplaces assigned to the selected cost center. You can also use wildcards.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected company. You can also use wildcards.

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected evaluation/report group.

Responsibility area

This selection criterion refers to the responsibility area in the workplace/machine master. Note: The user can only view those machines included in the responsibility areas assigned to the user.

Type

Selects the category of the machine/workplace displayed in the evaluation/report. You can select E (individual workplaces) and G (group workplaces).

Type

Selects the workplace type. You can select the following workplace types:

- P Workplace
- N Machine
- J Machining center
- L Line
- A Aggregate
- C CAQ inspection station
- R Reel-based manufacturing
- S Cutting unit

Show comments

If you select the checkbox *Show comments*, the table also shows entered comments.

Comment

If the input field *Comment* includes a text, the table only shows the data records that include this text as a comment. You can use * as a wild card. Please note case sensitivity.



You cannot use this selection field to search BDE comments.

Machine statuses > X minutes only

This parameter only refers to events of the type "machine status". The application will show the machine status if the posted time is greater than the entered value.

Event type

You can restrict the displayed events. The application shows all events, in case you have not restricted the selection.

Designation	Acronym
Machine status	M_MST
Production lock	M_PSPERRE
Operation postings	A_ADE
Personnel postings	P_ADE
Target value changes	M_VORGABE
Maßnahme	R_MASSNAHME
Resource posting	R_MELDUNG
Release of resource	R_FREIGABE
Resource status	R_STATUS
Maintenance reset	R_WART_RESET
Exceeding of maintenance	R_WART_EXCEEDED
DNC Upload	R_UPLOAD
DNC Download	R_DOWNLOAD
Transfer posting of resources	R_UMBUCHUNG
Beginning of status end of status	RES_STB RES_STE
BDE comment	HY_BEM: Display of BDE comments that have been entered in reference to an operation.

Please note: Posting of events depends on the customer's system and its use. Consequently, it might be the case that not all events listed here are relevant.

Date from ...to (shift/ time)

Use the date selection to restrict the period of time for the data you want to evaluate.

When selections are made using shift(s), the shift date is evaluated. If no shift is selected, all shifts are used.



Note that selection by shift is not supported for all event types. You can find detailed information on the shift selection [here](#).

If you select by time, the selection is based on the start date. Both times refer to the beginning or end of the date period specified above.



You can only evaluate *Group workplaces* if you select by *Time*. If you select by *Shift*, no data will be displayed because group workplaces do not refer to shifts.

Order / Article / MES order number

You can use these criteria to search for BDE postings:

- Log on OP, interrupt OP, log off OP, enter part quantities
- Log on staff, log off staff
- Change partitioning, change target cycle
- BDE comment

Machine history detail application

The machine history lists all events, such as status changes, order or personnel postings of a machine that occurred on the day. These have to be evaluated or listed in a shift of this day. The evaluations/reports show the following postings:



Postings based on machines/workplaces:

Postings for machine statuses recorded automatically (with direct machine connection)

Postings assigned manually at the terminal

Setting the production lock or changing default values relating to machines/workplaces (target cycle, partitioning) at the terminal

Automatic assignment of default values with operation postings



Postings based on orders:

Postings performed automatically (when shifts change)

Manual postings (logon, logoff, interruption) at the terminal.

The corresponding order is displayed additionally. If it is a manual posting, the person who did the posting is shown as well. If waiting period processing is active, the displayed logon time of the order represents the time of entry and may deviate from the point in time indicated in the order log record. →→→

**Postings based on staff:**

Automatic (when shifts change)

Manual logon or logoff processes of staff at the terminal

In addition, the application shows the corresponding personnel number and the operation for which the person produces.

**Postings based on resources:**

Machine postings resulting from the HYDRA Tool and Resource Management module (HYDRA-WRM), e.g. the application also shows exceeded maintenance activities or measures/comments.

**Information**

Shows BDE comments entered via the AIP terminal and stored with the operation.

The event "information" also shows the total duration of the respective status / event. The duration is always zero when a person or OP is logged on. The duration states the interval between the logging on and logging off if you interrupt/log off an OP or person.

Field description

The following paragraphs describe the data available in the table. It might be the case that the application does not show this data by default. Use the column selection function to add the required data.

Field description *workplace* category

Workplace

Workplace the event refers to.

Field description *event* category

Type

Image display of the type

Event type

Assign the recorded event. Possible values: see event

Event

Classifies the event collected at the machine in the table row. In the columns "Selection by shift" and "Selection by time" you can see events available for a specific selection.

Event type	Event	Selection by shift	Selection by time
Machine status	Machine status according to configuration Coloring is set according to the settings in the status text	Yes	Yes

Event type	Event	Selection by shift	Selection by time
	configuration..		
Production lock	Production lock set manually Production lock canceled manually	No	Yes
Operation postings	OP logged on OP interrupted OP logged off	Yes	Yes
Personnel postings	Person logged on Person logged off	Yes	Yes
Target value changes	Change partitioning/change target cycle	No	Yes
Exceeding of maintenance	Maintenance cycle exceeded	No	Yes
Maintenance reset	Maintenance reset	No	Yes
Information	BDE comment entered	Yes	Yes
Beginning of status end of status	Event and coloring according to configuration	No	Yes

Datum

Entry date of the event

Time

Entry time of the event

Duration

Time between the last event of this kind and the one currently displayed. The duration is only shown for the events "OP INTERRUPTED", "OP LOGGED OFF", "PERSON LOGGED OFF" as well as for machine statuses. In any other case, 0 is shown. These durations are synchronized with the BDE shift calendar, i.e. shift breaks are not included. Consequently, this value does not necessarily correspond to the period of time between logon and logoff.

Field description *master data* category

Workplace

Unique ID defined in the workplace configuration.

Designation

Machine name as defined in the workplace configuration.

Comment

Comment on the machine as defined in the workplace configuration.

Group

Capacity group which the machine was assigned to.

Cost center

Cost center as defined in the workplace configuration.

Company

Company as defined in the workplace configuration.

Responsibility area

Responsibility area required to view this workplace as defined in the workplace configuration.

Field description *order* category**Order type**

Order type of the order for which the event was collected.

order

Order number of the OP for which the event was recorded.

Sequence

Sequence number of the OP (provided that sequences are used).

OP

Operation number

Split

Split number of the operation (if split OPs are used)

SOP

Sub operation number (reserved).

Article

Article number produced by the operation; taken over from operation data.

Article designation/name

Article name of the article.

Field description *person* category**Person**

Personnel number of the person that has been logged on or off (only for Pers. postings)

Last name

The person's last name who was logged on or off (for personnel postings only).

First name

The person's first name who was logged on or off (for personnel postings only).

Name

Full name (last name, middle name and first name) of the person who was logged on or off (for personnel postings only).

Field description *status* category

If the event is a machine status, then this category shows the status number and status text name. This category shows the resource status for events based on resources.

Status

Status number of the assigned status

Status text

Status text of the assigned status

Receiving storage location

Destination when entering a resource status change (RES_STATUS).

Field description *maintenance* category

Maintenance type

Type of the maintenance

T: based on cycles,

B: based on operating hours

Z: based on time

Maintenance

Maintenance short text

Target cycles

For maintenance type T only: number of cycles until the maintenance is due again.

Actual cycles

For maintenance type T only: number of cycles accrued since resetting the maintenance interval. Value results from the machine data collection (MDE).

Planned hours of operation

For maintenance type B only: number of operating hours until maintenance falls due again.

Actual hours of operation

For maintenance type B only: number of operating hours accrued since resetting the maintenance interval. Value results from the machine data collection (MDE).

Next date

For maintenance type Z only: time when the maintenance falls due the next time.

Processing mode

For maintenance events (RES_WART):

R = Reset

Z = Threshold exceeded

A = Enabled/disabled

For changed resource statuses (RES_STATUS):

S = Change over status

Threshold 1 (in %)

Threshold until reaching due date

Threshold 2 (in %)

Threshold until reaching due date

Threshold 3 (in %)

Threshold until reaching due date

Active

“Active” flag of the maintenance activity at the time of the event.

Active (so far)

Only relevant for processing mode A: previous “active” status of the maintenance activity at the time when the maintenance activity was activated/deactivated.

Modified by

Editor who edited/set/reset the maintenance.

Datum

Date of editing/resetting

Time

Time of editing/resetting

Field description *measure* category

Maßnahme

Measure name

Designation

Name/description (long text) of the measure.

Reporting person

Person who created the measure.

Verantwortlicher

Person who has to carry out the measure.

Date of solution

Date when the measure has to be completed.

Priority

Priority of the measure.

Done

Flag indicating that the measure has been completed.

Done by

Person who marked the measure as being completed.

Field description *upload/download* category

(Not supported)

Field description *comment* category

Comment

Comment on the event entered by the employee.

Field description *changed partitioning* category

Partitioning

Partitioning

Cavity

Cavity number.

Type of modification

Reduced partitioning or increased partitioning.

Reason for change

Number of the reason for change.

Text of reason for change

Text of reason for change

Toolbar



Generate order (function authorization wphigenorder)

Use the "Generate order" function to create orders from work plans based on [Configuration](#).



Order information (function authorization: orin)

Request [Order information](#).

4 Machine Time Profile

Overview

Menu	Production facility/Resource management → Resource analysis → Machine time profile
Transaction code	mtpf
Function authorization	mtpf

Purpose

The machine time profile is the ideal tool for every planner, shift manager and production manager and is a report/evaluation of the production facility management function.

Integration

The machine time profile is used to visualize the production and downtime behavior of the machines in the foreman's area over a specified period. A clear, graphic bar chart shows which machine conditions were recorded at what time.

Selection criteria

The application provides the following selection criteria:

Workplace from ... to ...

This selection criterion refers to the workplace in the machine or workplace master data. You can also use wildcards (placeholders *).

Group from ... to ...

This selection criterion refers to the group in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected group. You can also use wildcards.

Report group

This selection criterion refers to the report groups. The application shows all workplaces/machines assigned to the selected evaluation/report group.

Responsibility area

This selection criterion refers to the responsibility area within the workplace/machine master. Note: The user can only view those machines that are included in the responsibility areas assigned to the user.

Short name

This selection criterion refers to the short name of machines in the master data. The application shows all machines or workplaces matching the entered character string. You can also use wildcards.

Company

This selection criterion refers to the company defined in the machine or workplace master data. The application shows all workplaces/machines assigned to the selected company. You can also use wildcards.

Status text

By entering a status text or a part of a status text, only those machines and workplaces are displayed that match the entered status text or the specified character string.

Status longer than x minutes

This selection criterion refers to the displayed statuses of the machines or workplaces. The graphic view only shows the statuses that were active at the machine longer than the specified period (in minutes).

Date from ...to (shift/ time)

Use the date selection to restrict the period of time for the data you want to evaluate.

When selections are made using shift(s), the shift date is evaluated. If no shift is selected, all shifts are used. You can only select machine and order data by shift.

If you select by time, the selection is based on the start date. Both times refer to the beginning or end of the date period specified above.



You can only evaluate *Group workplaces* if you select by *Time*. If you select by *Shift*, no data will be displayed because group workplaces do not refer to shifts.



The display shows the evaluation of the selected period of time whether the data is already archived or not.

Designation (name)

This field refers to the name of machines and workplaces defined in the machine master data. The application only shows the machines matching the specified character string. You can also use wildcards (placeholders *).

Cost center

This selection criterion refers to the cost center stored in the machine and/or workplace master data. The application shows all machines and/or workplaces assigned to the selected cost center. You can also use wildcards.

RPA number (Resource Performance Account)

By selecting one or more RPA accounts, the system only displays the statuses assigned to the RPA accounts or the status time entered for the RPA accounts in the graphical evaluation.

Display order

By activating this option, the current operations per machine and the individual statuses are displayed in the Gantt chart.

Show all machines

By default, the *Machine time profile* only displays machines for which the system recorded data in the selected period (and according to the further selection parameter).

If you check this option, you can display all machines, regardless of whether the system recorded data for the machines or not. If the system didn't record any data, the machine row is empty.

Show blocked machines

You can configure machines in the [Workplace and resource configuration](#) as *Blocked*. The display of the blocked machines in the *Machine time profile* depends on how the checkbox is set.

☐ Blocked and not blocked machines are displayed (default).

☐ Only machines are displayed that are not blocked.

☒ Only machines are displayed that are blocked.

If several selection criteria are used, overlapping results are displayed in the workplace overview.

View criteria

In addition to selecting data in the selection criteria, the graphic display may be changed by further view criteria:

General

In the *General* tab, you can group data for the display. Here you can specify a grouping option next to the option that a grouping should take place. The following groupings are possible at the moment:

- Group
- Cost center
- Company

Time scale

A drop-down box allows for the displayed scale to be divided into the dimensions *Seconds*, *Minutes*, *Hours*, *Days*, *Weeks* and *Months*. The scale is displayed in the selected dimension. The checkbox *Fit time scale into visible area* reduces or increases the selected time range in order for it to fit into the application (without scrolling). The + and - buttons allow for the data to be increased or reduced manually or step by step.

Workplace table

This multi-select box allows for the displayed data to be selected in the left table view. The following information is provided:

- Workplace
- Short name
- Rate of capacity utilization
- Reserve of rate of capacity utilization
- Cost center
- Group

Color status

In this tab, the displayed bar colors may be selected according to the RPA colors, status colors and colors for production and downtime.

A display according to the RPA colors is as follows:

RPA	Abbreviation	Designation (name)	Color
1	SUT	Secondary utilization time	Dark green
2	DCI	Disturbance-caused interruption (= technical interruption)	Red
3	LCI	Logistics-caused interruption (= organizational interruption)	Fuchsia
4	SCI	Staff-caused interruption	Purple
5	IMN	Idle mode, not scheduled	Black
6	IMS	Idle mode, scheduled	Dark gray
7	SET	Setup	Light turquoise
8	STA	Startup	Light blue
9	U8	Free (e.g. pilot production, or similar)	Dark blue
10	U9	Free (off work)	Brown
11	MUT	Main utilization time; "Production"	Light green
12	BKS	Neutral times, e.g. off, breaks etc., i.e. times that are not recorded	Olive

Operation colors

If data is displayed according to the selection criteria, it may be shown in different colors according to the following criteria:

- Category
- Order type
- Order
- Article
- Tool

Detail application *Machine time profile*

The *Machine time profile* is displayed and divided into a tabular view of the selected workplaces/machines and the graphic view of status development.

Tabular overview

The table overview shows the workplaces/machines including additional information, which have been selected for the graphic view. The type and grouping of data may be determined as described in the display criteria.

In addition to displaying master data for the selected workplaces/machines, such as short name, cost center and group, it is also possible to display the rate of capacity utilization and the reserve for the capacity utilization rate.

Rate of capacity utilization

The rate of capacity utilization is calculated from the ratio production time and total time.

Formula:

The rate of capacity utilization is always calculated on the basis of all downtimes (not only the ones displayed) compared to the total time.

Reserve for the rate of capacity utilization

The reserve for the rate of capacity utilization is calculated from the ratio of the displayed downtimes to the total time

Gaps which are smaller than the typical part running time on a machine and the required times (status *Assembly*), do not constitute a utilization reserve. These times are often not included and are therefore hidden.

$$\text{Formula: } \frac{\sum_{n=1}^{10} RPA_n - (\text{hidden downtimes})}{\sum_{n=1}^{11} RPA_n}$$

Formula:

Table view context menu**Workplaces/machines**

Opens the [Workplaces](#) application using the tabular overview of the workplaces/machines.

Status report

Opens the [Status report \(machine-related\)](#) application using the tabular overview of the workplaces/machines.

Graphic view

The graphic view shows which machine conditions were recorded at the individual machines at which point in time. The machine time profile has been designed to represent the production and downtime performance of machines of the foreman area over a specified period of time.



In case of very short status durations, it might happen - depending on the Gantt or screen resolution - that one pixel represents several seconds. For this reason, individual statuses might be displayed or hidden. For the display of very short statuses, you must increase the resolution.

"Graphic view" context menu**Order overview**

Opens the [Order overview](#) application using the operations displayed in the graphic view.

Operation overview

Opens the [Operation overview](#) application using the operations displayed in the graphic view.

Machine-related postings

Change to [Machine-related postings](#) with the transfer of the following parameter in the selection area:

- Machine number
- Date from - to
- Shift number



Note on the display of shifts: If a machine status is applied over several shifts, a machine posting is created in the system for each shift. In this case only one status is displayed for the machine time profile.

5 Cycle Parameters

Summary

Menu	Master data → Workplaces/ machines → Cycle parameters
Transaction code	cycpa
Function authorization	mdcycl

Usage

HYDRA provides the ability to monitor cycle times within the machine data recording function without requiring HYDRA process data processing to be used.

The purpose of this function is to configure the action and tolerance limits.

Integration

Values are marked in different colors in the *Workplace overview* depending on whether the action or tolerance limit was exceeded:

Standard	Black
If the value drops below or exceeds the action limit	Blue
If the value drops below or exceeds the tolerance limit	Red



If a limit is exceeded, no further processing steps are taken in HYDRA.

Requirement

Before defining any configurations, you must first set up the [machine](#).

Selection criteria

The application provides the following selection criteria:

Machine

Selection by machine/ workplace

Field descriptions

Machine

Machine for which the configuration applies.

Tolerance limit positive, negative

Values may not drop below or exceed the percentage values defined here. The cycle time of the logged on operation is always used as the target value for cycle time monitoring. This can be corrected at the terminal. The limit value is entered as a percentage of the target value.

Example: Target value: 20 sec/ cycle
 tolerance positive: 10 %
 tolerance negative: 5 %

Thus, this results in the following limit values:

 Upper limit value: 22 sec/ cycle
 lower limit value: 19 sec/ cycle

Action limit positive, negative

Percentage values can be entered here, triggering a warning once they have been reached. This is why the action limits should be defined more narrowly than the tolerance limits. The limit value is entered as a percentage of the target value.

6 Cycle progression

Overview

Menu	Resource management → Key figures → Cycle progression
Transaction code	Cycl
Function authorization	Cycl

Purpose

The purpose of this overview is to show a timely presentation of a machine's cycle development over a period of time that you can select.

Integration

The data displayed here are collected and saved as part of the machine data collection (MDE). Please also note the information about the database at the end of this section.

Requirements

Please also note the information about the database at the end of this section.

Selection criteria

The application provides the following selection criteria:

Workplace

Enter the number of the workplace/ machine, for which you would like to display a cycle progression.

Point in time ... to ...

When the application is pulled up, the point in time is predefined as follows:

From: Date = "yesterday"/ time = "now"

To: Date = "today"/ time = "now"

Choose the point in time for which the cycle progression should be displayed. Please keep in mind that the length of the point in time will affect data calculation and therefore the response time for the evaluation.

Grid

When the application is pulled up, the grid "point in time" is predefined. Choose the desired grid in which you would like the evaluation to run.



If grid spacing is chosen (not equal to "point in time"), then the calculated actual cycle is the arithmetic mean value of all random samples of actual cycles in the relevant grid spacing period:

$$\overline{IZY} = \frac{1}{n} \sum_{i=1}^n IZY_i$$

The time for the values in this case is the end time of the grid interval. Thus, for example, for hour grids, the values between 13.00 and 14.00 are averaged and 14.00 is displayed as the point in time.

Tabular report

Different presentation options can be chosen for the table view. The below-mentioned data is shown:

Date, time

Point in time when actual cycle data was saved. Please also note the information about the data basis at the end of this section.

Sec/ cycle (depending on the selected table)

Shows the actual cycle in [seconds/cycle].

Cycle/ sec (depending on the selected table)

Shows the actual cycle in [cycles/seconds].

Min/ cycle (depending on the selected table)

Shows the actual cycle in [minutes/cycle].

Cycle/ min (depending on the selected table)

Shows the actual cycle in [cycles/minutes].

LTL (lower tolerance limit)

Calculated lower tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: LTL = Target cycle - (Target cycle * [Tolerance limit, negative] / 100)

LAL

Calculated lower action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: LAL = Target cycle - (Target cycle * [action limit, negative] / 100)

UAL

Calculated upper action limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: $UAL = \text{Target cycle} + (\text{Target cycle} * [\text{action limit, positive}] / 100)$

UTL (upper tolerance limit)

Calculated upper tolerance limit for the selected machine based on the target cycle available when the actual cycle was saved and on the configuration [Cycle parameter](#).

Formula: $UTL = \text{Target cycle} + (\text{Target cycle} * [\text{Tolerance limit, positive}] / 100)$



The application does not show the target cycle that is active when saving an actual cycle.

Changes to data in the "machine-related postings" application do not affect this application.

Graphic detail applications

Similar to the tabular detail applications, there are four different detail applications available to show the values as a graphic display, each of which present the data in a different unit:

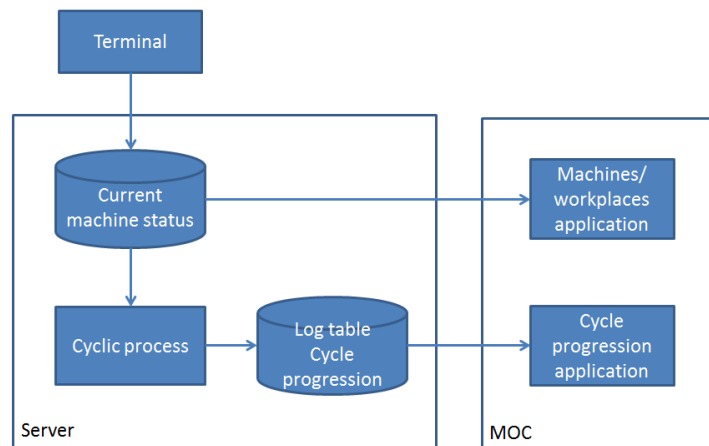
- Seconds/ cycle
- Cycles/ second
- Minutes/ cycle
- Cycles/ minute

The tolerance limits (red) and action limits (yellow) are shown in graphics.

Notes on the data basis for the display of cycle progression

The current actual cycle for each of the separate machines is stored together with the current point in time and the currently set target cycle in a special log table using a cyclic process.

Schematic process:



By default, the cycle for which the process stores the data in the log table for cycle progression is set to every 30 minutes. If necessary, this cycle can also be set to lower intervals (e.g. every 15 minutes) while taking into account the total capacity of the customer's system.



If the value of an actual cycle stays for a longer time at e.g. 0, the scheduler continues to create cyclic entries. All entries include the same actual cycle and the same time stamp (time stamp when the value was set).

Subsequent changes to data in the "machine-related postings" application do not affect the cyclic process.

The application *Cycle progression* accesses values stored in the log table and displays these as a graph in the time progression.

By default, the cycle data for a machine are available for 50 calendar days. If necessary, the data for each machine can also be stored for a longer time (e.g. 90 days) while taking into account the total capacity of the customer's system (must be assured by the customer).

For both cases (modifying logging interval or availability duration of the data), the respective entry must be adjusted in the [Scheduler](#):

Field	Value
Type	S (Standard)
Category	I (interval)
Alterable	Yes
Visible	Visible
Product key	MDE-BP

Field	Value
License key	MDE-BP
HYDRA users	0
Command	./mz_zykl.exe 50
Comment	MDE cycle progression
Interval	00:30:00
Active	☒

You must restart the MES after having modified the values.



The value behind the command is the availability period of log data in days for cycle analyses.